

Au-23-1(a)

Components of context-free grammar :

1. V is a finite set called the variables. Also called N i.e. non-terminals.
2. Σ is a finite set, disjoint from V also called T i.e. terminals.
3. R or P is a finite set of rules or productions.
4. S is the start variable, $S \in V$

Yes, there are few components that have similarities with any components of regular language or finite automata.

1. Terminals are similar with the alphabet. Also both are finite set
2. Productions or rules are similar with the transition function.
3. Start variable is similar in concept to the start state.

a. Derivation: The sequence of substitution to obtain a string is called derivation

Parse Tree: It is a tree like structure where each node represent terminal or non terminal symbol and edge depict the sequence of substitution.

Difference between derivation and parse tree:

Derivation are represented as a linear sequence of production rules.

parse tree is a graphical structure represent the hierarchical relationship between symbols.

b. Leftmost derivation: Applying production to the leftmost variable in each step

Rightmost derivation: Applying production to the rightmost variable in each step.

Difference between leftmost and rightmost derivation

Both derivation result in the same string but they differ in the order in which non terminals are expanded during the derivation process.

Ambiguity: In context free grammar ambiguity refers to a situation where a grammar generates the same string in several different ways.

Given grammar:

$$S \rightarrow AB$$

$$A \rightarrow aA \mid abA \mid \epsilon$$

$$B \rightarrow bB \mid abB \mid \epsilon$$

$$S \rightarrow AB$$

$$\rightarrow abAB \quad [A \rightarrow abA]$$

$$\rightarrow abAabB \quad [B \rightarrow abB]$$

$$\rightarrow ababB \quad [A \rightarrow \epsilon]$$

$$\rightarrow abab \quad [B \rightarrow \epsilon]$$

$$S \rightarrow AB$$

$$\rightarrow aAB \quad [A \rightarrow aA]$$

$$\rightarrow aB \quad [A \rightarrow \epsilon]$$

$$\rightarrow abB \quad [B \rightarrow bB]$$

$$\rightarrow ababB \quad [B \rightarrow abB]$$

$$\rightarrow abab \quad [B \rightarrow \epsilon]$$

The given grammar is ambiguous because the string "abab" derived ambiguously in that grammar.

Given grammar,

$$S \rightarrow SSx \mid SSy \mid SSz \mid a \mid b \mid c$$

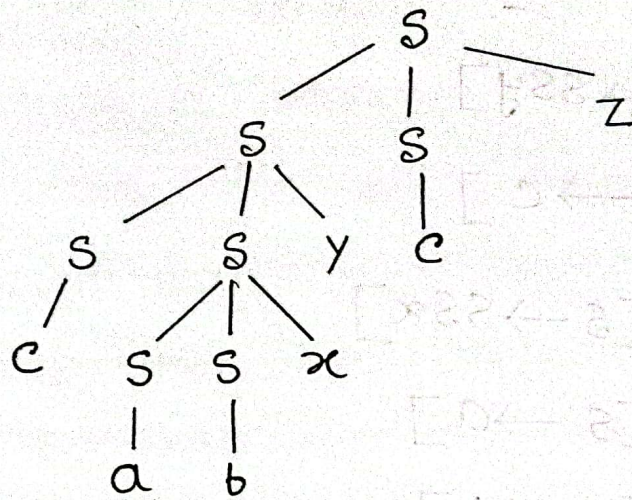
Left most derivation:

$s \rightarrow SSz \quad [s \rightarrow SSz]$
 $\rightarrow SSySz \quad [s \rightarrow SSy]$
 $\rightarrow cSySz \quad [s \rightarrow c]$
 $\rightarrow cSSxySz \quad [s \rightarrow SSx]$
 $\rightarrow casxySz \quad [s \rightarrow a]$
 $\rightarrow cabxySz \quad [s \rightarrow b]$
 $\rightarrow cabxyz \quad [s \rightarrow c]$

Rightmost derivation:

$s \rightarrow SSz \quad [s \rightarrow SSz]$
 $\rightarrow scz \quad [s \rightarrow c]$
 $\rightarrow SSycz \quad [s \rightarrow SSy]$
 $\rightarrow SSSxyz \quad [s \rightarrow SSx]$
 $\rightarrow SSbxyz \quad [s \rightarrow b]$
 $\rightarrow sabxyz \quad [s \rightarrow a]$
 $\rightarrow cabxyz \quad [s \rightarrow c]$

Parse tree for the string:



Ans-23-24) 2(a) or

a) 0^*1^*

CFG:

$S \rightarrow 0S1 \mid \epsilon$

b) $1(01)^*$

CFG:

$S \rightarrow 1A$

$A \rightarrow 01A \mid \epsilon$

c) $(1100)^*$

CFG:

$S \rightarrow 11S \mid 0S \mid \epsilon$

Au-23-2(b)

$$i) L = \{a^{2n} b^m c^n \mid n \geq 0\}$$

$$A \rightarrow aaAc \mid B \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

$$ii) L = \{a^m b^n c^{m+n} \mid n \geq 0\}$$

$$A \rightarrow aAc \mid B \mid \epsilon$$

$$B \rightarrow bBc \mid \epsilon$$

Au-23-2(c)

$$R \rightarrow asa \mid bRb \mid S$$

$$S \rightarrow aTb \mid bTa \mid aS$$

$$T \rightarrow XT \mid X \mid \epsilon$$

$$X \rightarrow a \mid b$$

Step-2.1: Make a new start variable

$$R_0 \rightarrow R$$

$$R \rightarrow aSa | bRb | S$$

$$S \rightarrow aTb | bTa | aS$$

$$T \rightarrow xTx | x | \epsilon$$

$$x \rightarrow a | b$$

Step-2: Remove null production

$$R_0 \rightarrow R$$

$$R \rightarrow aSa | bRb | S$$

$$S \rightarrow aTb | bTa | aS | ab | ba$$

$$T \rightarrow xTx | x | xx$$

$$x \rightarrow a | b$$

Step-3: Remove unit rules

$$R_0 \rightarrow aSa | bRb | aTb | bTa | aS | ab | ba$$

$$R \rightarrow aSa | bRb | aTb | bTa | aS | ab | ba$$

$$S \rightarrow aTb | bTa | aS | ab | ba$$

$$T \rightarrow xTx | a | b | xx$$

$$x \rightarrow a | b$$

Step-4]

$R_0 \rightarrow XSX | XRX | XTX | XTX | XS | XX | \epsilon XX$

$R \rightarrow XSX | XRX | XTX | XTX | XS | XX | XX$

$S \rightarrow XTX | XTX | XS | XX | XX$

$T \rightarrow XTX | a | b | xx$

$x \rightarrow a | b$

Step-5]

$R_0 \rightarrow XSX | XRX | XTX | XS | XX$

$R \rightarrow XSX | XRX | XTX | XS | XX$

$S \rightarrow XTX | XS | XX$

$T \rightarrow XTX | a | b | xx$

$x \rightarrow a | b$

Step-6]

$R_0 \rightarrow XV_1 | XV_2 | XV_3 | XS | XX$

$R \rightarrow XV_1 | XV_2 | XV_3 | XS | XX$

$S \rightarrow XV_3 | XS | XX$

$T \rightarrow XV_3 | a | b | xx$

$V_1 \rightarrow SX$

$V_2 \rightarrow RX$

$V_3 \rightarrow TX$

$x \rightarrow a | b$

This is the required chomsky normal form

Au-23-3(a)

We remove an ϵ rule $A \rightarrow \epsilon$ where A is not the start variable. Then for each occurrence of an A on the right-hand side of a rule, we add a new rule with that occurrence deleted. If $R \rightarrow uAv$ then we add rule $R \rightarrow uv$. So for rule $R \rightarrow uAvAw$ we add $R \rightarrow uvAw$ and $R \rightarrow uAvw$ and $R \rightarrow uvw$. If a rule $R \rightarrow A$ then we add $R \rightarrow \epsilon$.

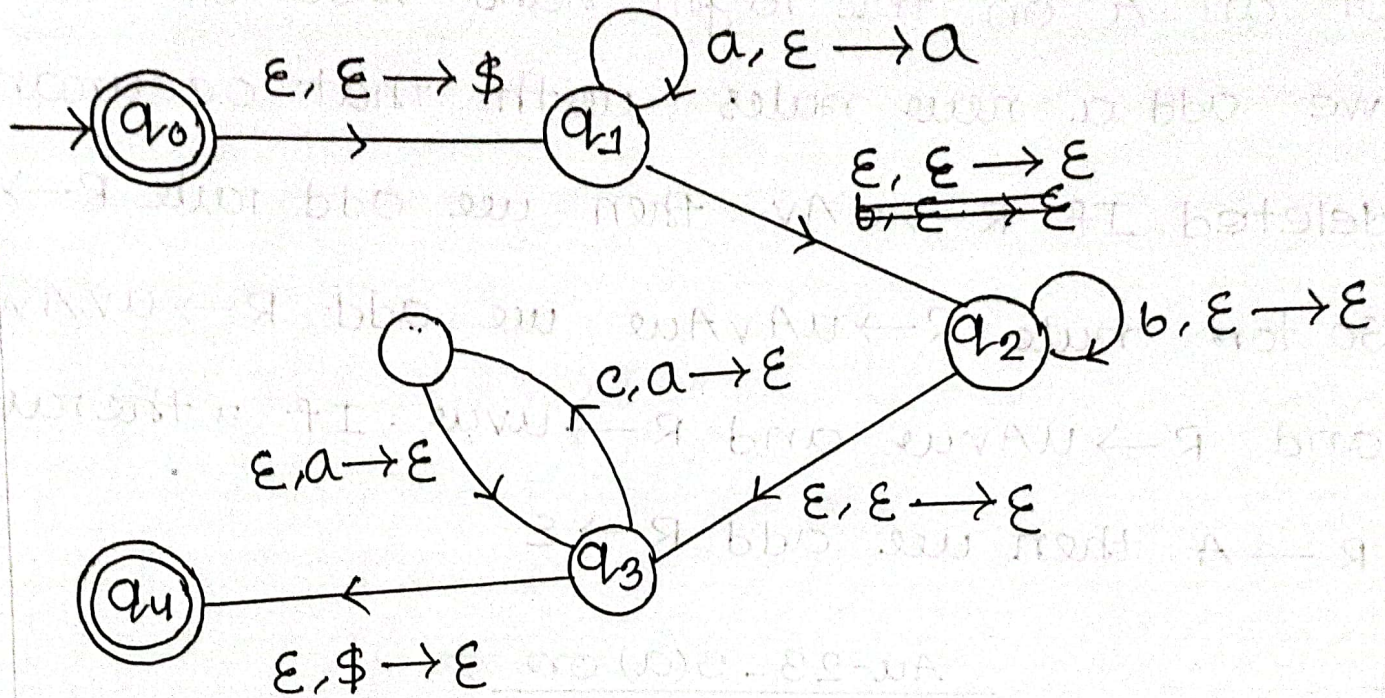
Au-23-3(a) on

Pushdown automata are more powerful than finite automata because it has an extra component called a stack. The stack provides additional memory beyond the finite amount available in control. The stack allows pushdown automata to recognize some non-regular languages.

Au-23-3(b)

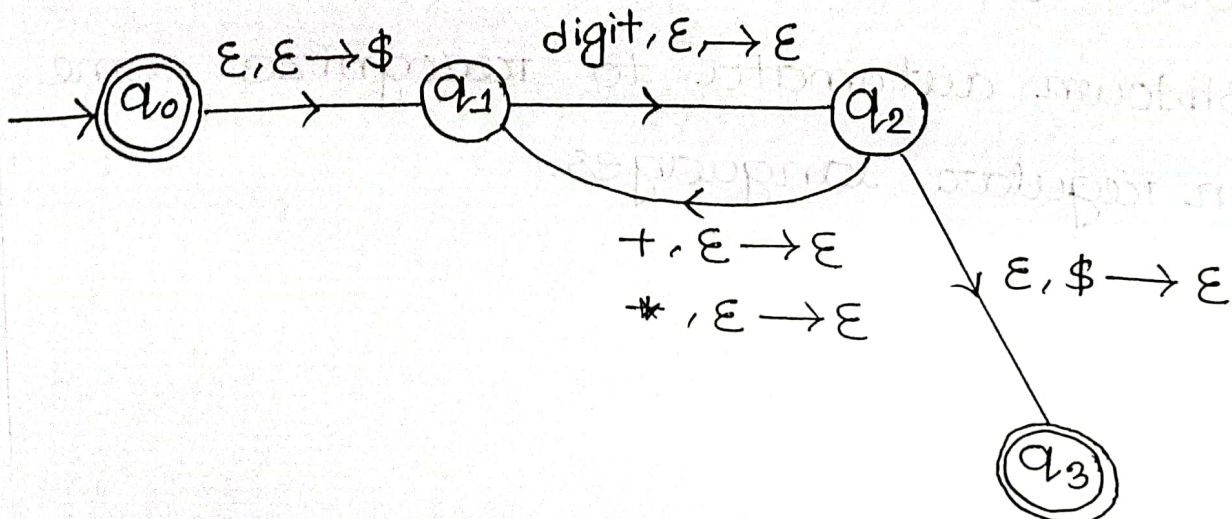
Given language: $L = \{a^{2n}b^m c^n \mid n \geq 0\}$

Pushdown automata:



Au-23-3(b) or

A pushdown automata that recognize any arithmetic expression involving $+, *$ and any one digit integer

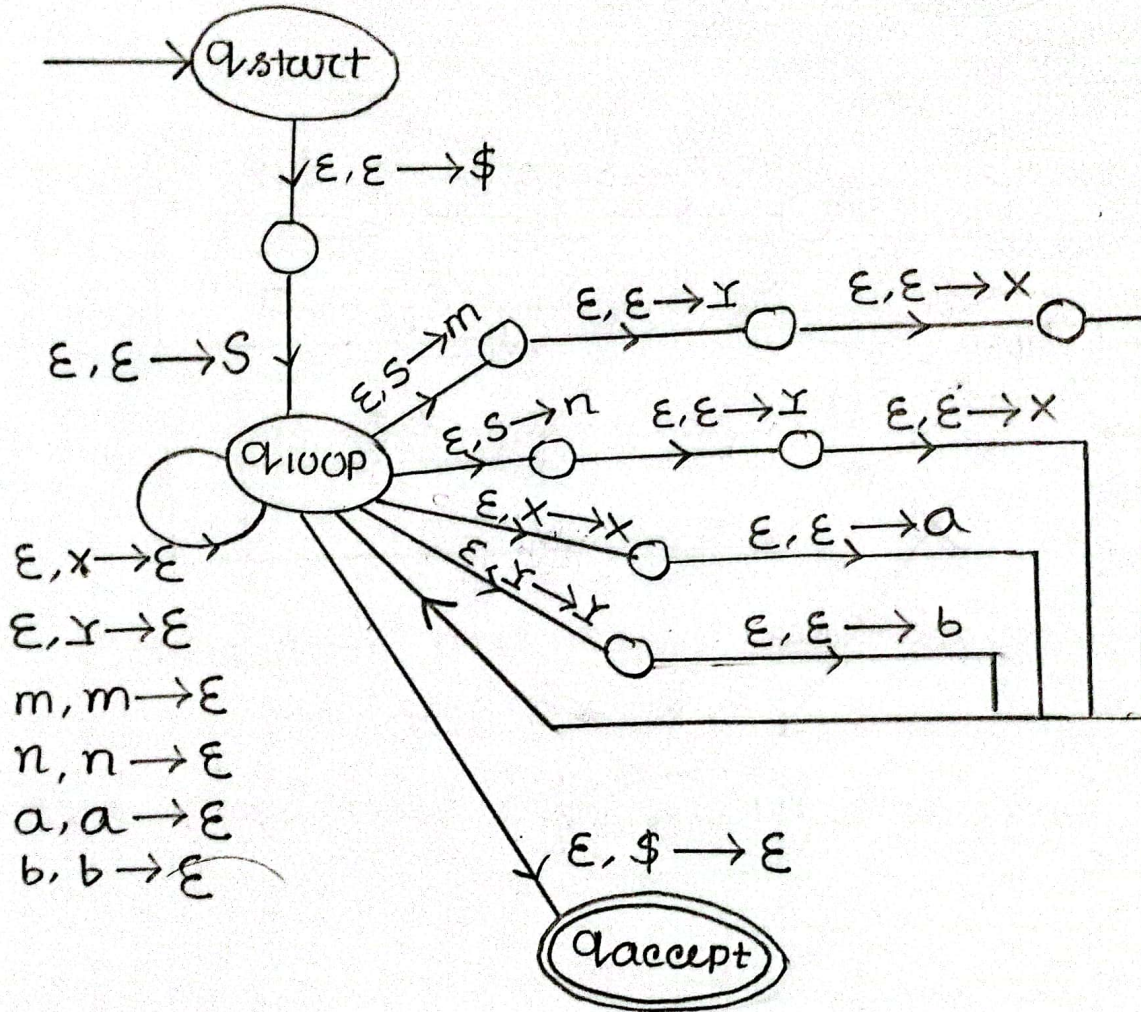


CFG to PDA

$S \rightarrow xym | xy n$

$x \rightarrow ax | \epsilon$

$y \rightarrow by | \epsilon$



$S \rightarrow axbY$

$x \rightarrow aYa| \epsilon$

$Y \rightarrow bxb|c| \epsilon$

